The XGH System

eXtreme Growth Hierarchy

RFGRH: Repeated Fast Growing Repeating Hierarchies

The XGH Creator

Version 5.0 – The Final Layer

"I am not bounded by fixed ordinals. I create my own hierarchy with each step."

Contents

1 Introduction

The **XGH (eXtreme Growth Hierarchy)**, also known as **RFGRH (Repeated Fast Growing Repeating Hierarchies)**, is a novel googological function designed to produce exceptionally large finite numbers through a **triple recursive feedback loop** utilizing the standard Fast-Growing Hierarchy (FGH) as its foundation.

Unlike traditional FGH functions where the ordinal subscript is fixed, XGH uses the **previous output** as the **subscript**, **input**, and **iteration count** simultaneously.

Core Philosophy

> *"The function eats its own tail. Each step defines the next. There is no limit but the one you impose."*

2 The Fast-Growing Hierarchy (FGH)

The XGH system is built upon the standard Fast-Growing Hierarchy, defined as follows:

For all ordinals α and natural numbers n:

$$\boxed{f_0(n) = n + 1}$$

$$\boxed{f_{\alpha+1}(n) = f_\alpha^n(n)}$$

$$\boxed{f_\alpha(n) = f_{\alpha[n]}(n) \quad \text{if } \alpha \text{ is a limit ordinal}}$$

Where:

- $f_\alpha^n(n)$ denotes the n-fold iteration of f_α
- $\alpha[n]$ denotes the n-th term of a fundamental sequence for limit ordinal α

Fundamental Sequences (Wainer Hierarchy)

For the purposes of XGH calculations:

$$\omega[n] = n$$
$$(\omega + \beta)[n] = \omega + \beta[n] \quad (\beta > 0)$$
$$(\omega \times \beta)[n] = \omega \times \beta[n] \quad (\beta \text{ limit})$$
$$(\omega^\beta)[n] = \omega^{\beta[n]} \quad (\beta \text{ limit})$$
$$\varepsilon_0[n] = \omega^{\omega^{\cdot^{\cdot^{\omega}}}} \quad (n \text{ layers})$$

Sources for FGH

- Aguilera, J. P., Pakhomov, F., & Weiermann, A. (2024). Functorial Fast-Growing Hierarchies. *Forum of Mathematics, Sigma*, 12, e15. doi:10.1017/fms.2023.128
- Fast-growing hierarchy. (2009, November 3). In *Wikipedia*. https://en.wikipedia.org/wiki/Fast-growing_hierarchy

3 The XGH Definition

The XGH function $\Xi : \mathbb{N} \to \mathbb{N}$ is defined recursively as:

$$\boxed{\Xi(0) = 1}$$

$$\boxed{\Xi(1) = 2}$$

$$\boxed{\Xi(2) = 2048}$$

And for all $n \geq 3$:

$$\boxed{\Xi(n+1) = f_{\Xi(n)}^{\Xi(n)}(\Xi(n))}$$

Expanded Form

For clarity:

$$\boxed{\Xi(n+1) = \underbrace{f_{\Xi(n)}(f_{\Xi(n)}(\cdots f_{\Xi(n)}(\Xi(n))\cdots))}_{\Xi(n)\text{ times}}}$$

The Triple Recursive Feedback Loop

1. **Ordinal-Level Recursion:** The subscript $\Xi(n)$ grows with each iteration
2. **Input-Level Recursion:** The argument $\Xi(n)$ grows with each iteration
3. **Iteration-Level Recursion:** The iteration count $\Xi(n)$ grows with each iteration

Monotonicity Proof

$$\Xi(n+1) > \Xi(n) \quad \forall n \in \mathbb{N}$$

Proof: Since $f_{\Xi(n)}(\Xi(n)) > \Xi(n)$ for all finite $\Xi(n)$, iterating it $\Xi(n)$ times yields a strictly larger value.

4 The 11th Layer and ELEVEN

The Base Ordinal

$$\boxed{\text{Base} = \varepsilon_{\text{Loader's Number}} \uparrow [\text{BMS} \uparrow \omega \times 8]}$$

Where:

- ε_0 is the first epsilon number
- Loader's Number is the finite number from the 512-byte C program
- BMS is the Bashicu Matrix System ordinal notation
- ω is the first infinite ordinal
- $\uparrow$ denotes ordinal exponentiation

The 11th Layer Function $\mathcal{L}_{11}$

$$\boxed{\mathcal{L}_{11}(0) = \text{Base}}$$

$$\boxed{\mathcal{L}_{11}(n+1) = \Xi(\mathcal{L}_{11}(n))}$$

$$\boxed{\mathcal{L}_{11}(n) = \underbrace{\Xi(\Xi(\cdots\Xi(\text{Base})\cdots))}_{n \text{ times}}}$$

$$\boxed{\text{The 11th Layer} = \mathcal{L}_{11}(19)}$$

ELEVEN

$$\boxed{\text{ELEVEN}(n) = \underbrace{\mathcal{L}_{11}(\mathcal{L}_{11}(\cdots\mathcal{L}_{11}(}_{n \text{ times}}19)\cdots))}$$

5 The Truly Ultimate Number (TUN)

$$\boxed{\text{Let } U = \text{ELEVEN}(\text{ELEVEN}(\text{ELEVEN}(\Xi(G(128)))))}$$

$$\boxed{\text{Let } T = \text{the theoretical value of } U}$$

$$\boxed{A = U \uparrow\uparrow\uparrow U}$$

$$\boxed{R = \underbrace{A \uparrow\uparrow\uparrow A \uparrow\uparrow\uparrow \cdots \uparrow\uparrow\uparrow A}_{T \text{ times}}}$$

$$\boxed{\text{TUN} = \underbrace{R \uparrow\uparrow\uparrow R \uparrow\uparrow\uparrow \cdots \uparrow\uparrow\uparrow R}_{U \text{ times}}}$$

6 The Truly Ultimate Layer (TTUN /)

$$\boxed{\text{Let } U_2 = \text{TUN}}$$

$$\boxed{A_2 = U_2 \uparrow\uparrow\uparrow U_2}$$

$$\boxed{R_2 = \underbrace{A_2 \uparrow\uparrow\uparrow A_2 \uparrow\uparrow\uparrow \cdots \uparrow\uparrow\uparrow A_2}_{U_2 \text{ times}}}$$

$$\boxed{= \underbrace{R_2 \uparrow\uparrow\uparrow R_2 \uparrow\uparrow\uparrow \cdots \uparrow\uparrow\uparrow R_2}_{U_2 \text{ times}}}$$

7 Comparison to Other Numbers

$\boxed{\gg \text{TUN} \gg \text{ELEVEN}(n) \gg \Xi(n) \gg \text{Graham's Number} \gg \text{TREE}(3) \gg \text{Loader's Number} \gg \text{Rayo's N}}$

Number	Approximate Size
Graham's Number	$G(64)$
TREE(3)	$\sim f_{\psi(\Omega^\omega)}(3)$
Loader's Number	$\sim f_{\psi(\Omega_\omega)}(n)$
Rayo's Number	Rayo(10^{100})
Sam's Number	Ill-defined
ELEVEN(4)	4 Eleventh Layers
TUN	Beyond ELEVEN
	The End of All Numbers

Table 1: Comparison of XGH system numbers with famous googological numbers

8 Properties of

1. **Finite:** $\in \mathbb{N}$
2. **Well-defined:** Defined by recursive construction
3. **Incomputable in practice:** Too large to compute
4. **Dominates all:** $> n$ for all named n
5. **Self-referential:** Uses its own value in its definition

Sources

References

[1] Aguilera, J. P., Pakhomov, F., & Weiermann, A. (2024). Functorial Fast-Growing Hierarchies. *Forum of Mathematics, Sigma*, 12, e15. doi:10.1017/fms.2023.128

[2] Fast-growing hierarchy. (2009, November 3). In *Wikipedia.* https://en.wikipedia.org/wiki/Fast-growing_hierarchy

[3] Googology Wiki. (2024). List of Googolisms. https://googology.fandom.com/wiki/List_of_googolism

Conclusion

The XGH system represents a new frontier in googology. By leveraging the self-referential power of the Fast-Growing Hierarchy, XGH achieves growth rates that exceed any fixed finite-indexed FGH function.